

EDUGENIE

Phase 4 – Project Planning

AI-Powered Learning Assistant

1. Project Planning Overview

This project plan defines the activities, sequence, milestones, resources, testing, documentation, and demonstration work required to develop the EduGenie AI-powered learning assistant. The plan follows the eight phases required for the project submission and provides a structured path from initial ideation through final demonstration.

2. Project Objectives

- Develop a functional Generative AI-powered educational assistant.
- Implement Question & Answer functionality.
- Implement topic explanation functionality.
- Implement text summarization functionality.
- Implement quiz generation functionality.
- Implement learning recommendation functionality.
- Provide a simple web interface for interacting with the application.
- Integrate the application with the Google Gemini API.
- Test the application features and document the results.
- Prepare the final project documentation and demonstration.

3. Project Development Phases

Phase	Phase Name	Main Activities	Expected Output
1	Brainstorming & Ideation	Identify the problem, brainstorm AI-based solutions, define the project idea and objectives.	Brainstorming & Ideation document
2	Requirement Analysis	Identify functional and non-functional requirements, users, inputs, outputs, and technology requirements.	Requirement Analysis document
3	Project Design	Design system architecture, use cases, application flow, and major components.	Design document and diagrams

4	Project Planning	Define tasks, sequence, milestones, resources, schedule, risks, and deliverables.	Project Plan
5	Project Development	Develop frontend, backend, AI integration, modules, configuration, and automated tests.	Working EduGenie application
6	Project Testing	Execute test cases, verify features, record results, and capture evidence.	Test cases, results, and testing report
7	Project Documentation	Prepare final report covering implementation, testing, results, limitations, and future work.	Final project report
8	Project Demonstration	Record the application demonstration and provide the public video link.	Demo video and public link

4. Task Breakdown

ID	Task	Description	Phase
T01	Project setup	Create the project repository and development structure.	Phase 5
T02	Backend setup	Configure FastAPI application and project settings.	Phase 5
T03	AI integration	Configure the Gemini client and environment variables.	Phase 5
T04	Q&A module	Implement educational question answering.	Phase 5
T05	Explanation module	Implement topic explanation.	Phase 5
T06	Summary module	Implement text summarization.	Phase 5
T07	Quiz module	Implement quiz generation.	Phase 5
T08	Learning path module	Implement learning recommendations.	Phase 5
T09	Frontend	Implement the web interface and client-side interactions.	Phase 5
T10	Testing	Run automated and manual tests and record evidence.	Phase 6

T11	Documentation	Prepare final project documentation.	Phase 7
T12	Demonstration	Record and publish the project demonstration video.	Phase 8

5. Suggested Project Timeline

The following schedule can be used as a practical academic project timeline. The exact dates can be adjusted to match the student's institutional submission schedule.

Week	Planned Work	Key Deliverable	Status
Week 1	Brainstorming & Ideation	Problem statement, ideas, objectives	Planned
Week 2	Requirement Analysis	Functional and non-functional requirements	Planned
Week 3	Project Design	Architecture and use-case diagrams	Planned
Week 4	Project Planning & Setup	Development plan and environment setup	Planned
Week 5	Backend & AI Integration	FastAPI backend and Gemini integration	Planned
Week 6	Feature Development	Q&A, explanation, summary, quiz, learning path	Planned
Week 7	Frontend & Integration	Integrated working web application	Planned
Week 8	Testing	Test cases and testing report	Planned
Week 9	Documentation	Final project report	Planned
Week 10	Demonstration & Submission	Demo video and final GitHub repository	Planned

6. Resources Required

- Computer or mobile device capable of accessing the development environment.
- Visual Studio Code or another suitable code editor.
- Python 3.11 or later.
- Internet connection for Generative AI API access.
- Google Gemini API key.
- Git and a public GitHub repository.
- Google Drive for the project demonstration video.
- Python packages listed in requirements.txt.

7. Planned Technology Stack

Technology	Role	Planned Use
Python	Programming language	Backend and application logic
FastAPI	Backend framework	REST API and web application backend
Google Gemini	Generative AI	AI-generated educational responses
HTML	Frontend	Application structure
CSS	Frontend	User interface styling
JavaScript	Frontend	User interaction and API calls
Pydantic	Validation	Request and response data validation
Pytest	Testing	Automated API testing
Git/GitHub	Version control	Project source management and submission

8. Risk Management

Risk	Possible Impact	Mitigation	Priority
Invalid or missing Gemini API key	AI features may not work	Use .env configuration and .env.example; verify the key before testing.	High
API service or network unavailable	AI responses may fail	Add error handling and perform testing with a stable internet connection.	High
Invalid user input	Unexpected API responses	Validate requests and provide clear input guidance.	Medium
Integration errors	Features may not work together	Test each module and then perform end-to-end testing.	High
Sensitive credentials committed to GitHub	API key exposure	Exclude .env through .gitignore and never publish real credentials.	High
Late documentation	Submission may be delayed	Prepare documentation alongside development and testing.	Medium

9. Testing Plan

Testing will be carried out after the main application features are implemented. The testing phase will include automated API tests and manual verification of the five primary learning features.

- Verify the application starts successfully.
- Verify the health endpoint responds correctly.
- Verify Q&A accepts a question and returns an AI response.
- Verify topic explanation produces an explanation.

- Verify summarization processes submitted text.
- Verify quiz generation produces the expected quiz structure.
- Verify learning recommendations are returned for a requested topic.
- Verify invalid or incomplete input is handled appropriately.
- Record pass/fail results and capture screenshots for important tests.

10. Deliverables

- Brainstorming & Ideation document.
- Requirement Analysis document.
- Project Design document and diagrams.
- Project Plan document.
- Complete EduGenie source code.
- Test cases and testing report.
- Final project documentation.
- Public GitHub repository.
- Project demonstration video hosted on Google Drive.
- Public demonstration video link.

11. Completion Criteria

The project will be considered ready for submission when:

- All five major EduGenie learning features are implemented.
- The frontend and backend communicate correctly.
- The Gemini API integration works with a valid API key.
- Automated and manual testing has been completed.
- Testing results have been documented.
- All required phase-wise documents have been prepared.
- The GitHub repository contains the required project files and is public.
- The .env file and real API key are excluded from the public repository.
- The demonstration video is uploaded to Google Drive with public viewing access.

12. Project Planning Conclusion

This project plan provides a structured approach for developing and submitting EduGenie across the required eight phases. Following the planned sequence will help maintain clear project progress, reduce integration and submission risks, and provide evidence for each stage from ideation through final demonstration.